

M^5 Day 14: More Exponentials

Name: _____

1. A jacket costs \$100. Calculate the price of the jacket in each of these scenarios:

- a. A single 50% discount is applied.

- b. A 20% discount is followed by a 30% discount.

- c. A 30% discount is followed by a 20% discount.

- d. Six successive discounts of 10% each are applied.

2. An amoeba doubles itself every 3 minutes and fills a container in 60 minutes. How long would it take to fill the container if you started with two amoebas instead?

3. Amazing facts about vampires:

- a. Exactly 400 years ago today, the first vampire mutated from a rabid bat that was struck by lightning.
- b. Vampires never die.
- c. Every 10 years, a vampire turns one human into a vampire by gorging on their blood.

If vampires exist, how many vampires would there be today? Do vampires exist?

4. An investment in the stock market gains 10% annually. This means that each year the account total is multiplied by a factor of 1.10. Which of these accounts will have more money?

- a. A \$1000 initial investment that is left alone to compound annually for 20 years.
- b. A \$2000 initial investment that is left alone to compound annually for 10 years.

M^5 Day 13: Sine & Cosine

Name: _____

1. How many radians does the minute hand of an analog clock sweep through between the times of 1:00pm and 1:45pm?

2. Evaluate the expression without using a calculator:

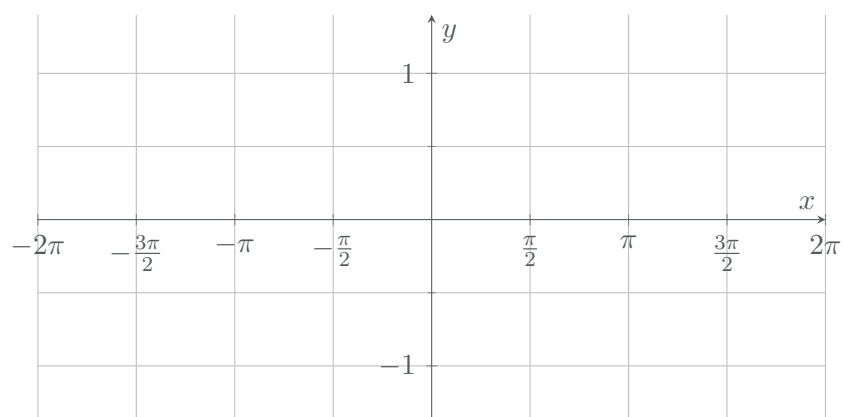
a. $\sin\left(\frac{\pi}{2}\right)$

b. $\cos\left(\frac{3\pi}{2}\right)$

c. $\sin\left(-\frac{\pi}{6}\right)$

3. Use the definition of $\cos$ to explain why $\cos(-x) = \cos x$ for all x .

4. Plot the graph of $\sin x$ below:



5. Plot the graph of $\cos x$ below:

